Cisco PE Router QoS Congestion Event SOP (KOR-PE-01, Koramangala Datacenter)

Node: KOR-PE-01 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_9540

Description

This Standard Operating Procedure (SOP) addresses the scenario where the Cisco PE router (KOR-PE-01) at the Koramangala Datacenter experiences a Quality of Service (QoS) congestion event, characterized by microbursts, interface tail drops, and QoS queue starvation. This failure domain arises due to the interaction of multiple protocol and hardware mechanisms. On Cisco equipment, QoS is implemented using the Modular QoS CLI (MQC) framework, which classifies and polices traffic based on predetermined criteria. However, when the QoS policy is not configured correctly or the interface is experiencing congestion, packets may be dropped or delayed, leading to microbursts and interface tail drops. Furthermore, if the QoS policy is not properly managed, it can result in QoS queue starvation, where high-priority traffic is not transmitted promptly, compromising network performance. The severity of this issue is classified as Warning (Alarm Code: ALARM_9540), necessitating prompt attention to prevent potential network disruptions.

Pre-requisite Checks

- check: Verify the Koramangala Datacenter is accessible and the KOR-PE-01 node is reachable via console or SSH.

- check: Check the redundant link status to ensure there are no issues with the primary or secondary links.

- check: Confirm that all QoS policies are correctly configured and not causing the congestion event.

Potential Root Causes

Cause	Probability
Incorrect QoS policy configuration	High
Interface congestion due to high traffic volumes	Medium
Hardware or software issues with the KOR-PE-01 node	Low

Diagnostic Commands & Expected Outputs

```
Command: show policy-map interface, Expected Output: Policy map is applied to the interface, and QoS policy is correctly configured., Description: This command displays the QoS policy map applied to the interface, allowing us to verify the correctness of the QoS configuration.

Command: show interface, Expected Output: Interface is operational, and no errors or warnings are reported., Description: This command provides an overview of the interface's operational status, including
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error and warning counts, which can help identify congestion or
hardware issues.

Command: debug policy-map, Expected Output: QoS policy is being
enforced, and packets are being classified and policed according to
the policy., Description: This command enables debugging for the
policy map, allowing us to observe the QoS policy in action and
identify any issues with packet classification and policing.
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Resolution Steps

Step 1: Step: Verify and correct the QoS policy configuration, Command: configure terminal policy-map class ...configuring QoS policy... end-policy-map, Description: Carefully review the QoS policy configuration and correct any errors or inconsistencies, ensuring that the policy accurately reflects the network's requirements and traffic profiles.

Step 2: Step: Adjust interface settings to reduce congestion, Command: interface ...configuring interface settings... end-interface, Description: Examine the interface settings and adjust them as necessary to reduce congestion and improve network performance, such as increasing the buffer size or modifying the congestion avoidance algorithm.

Step 3: Step: Upgrade or replace hardware or software components as necessary, Command: ...performing hardware or software upgrade or replacement... end-config, Description: If the root cause is identified as hardware or software issues with the KOR-PE-01 node, perform the necessary upgrades or replacements to rectify the problem.

Post-Check Verification

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Command: show policy-map interface, Expected Output: Policy map is
correctly applied to the interface, and QoS policy is properly
configured., Description: Verify that the QoS policy map is correctly
applied to the interface and that the QoS policy is properly
configured.

Command: show interface, Expected Output: Interface is operational,
and no errors or warnings are reported., Description: Verify that the
interface is operational and that no errors or warnings are reported,
indicating that congestion or hardware issues have been resolved.

Command: show policy-map, Expected Output: QoS policy is correctly
enforced, and packets are being classified and policed according to
the policy., Description: Verify that the QoS policy is correctly
enforced and that packets are being classified and policed according
to the policy, ensuring that network performance and quality are
maintained.
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Roll-back Procedures

- Step: Revert QoS policy configuration to previous version, Command: configure terminal policy-map class ...restoring previous QoS policy configuration... end-policy-map, Description: If the fix fails, revert the QoS policy configuration to its previous version to restore the original configuration.

- Step: Revert interface settings to previous configuration, Command: interface ...restoring previous interface settings... end-interface, Description: If the fix fails, revert the interface settings to their previous configuration to restore the original settings.

- Step: Revert hardware or software changes as necessary, Command: ...reverting hardware or software changes... end-config, Description: If the fix fails, revert any hardware or software changes made during the resolution process to restore the original configuration.